

Retrieval-Augmented Language Reasoning over Graph-Structured Data: A Report of GNN-Enhanced RAG Approaches

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Abstract

Large Language Models (LLMs) have demonstrated strong performance across a wide range of natural language processing tasks, yet they remain fundamentally limited when reasoning over structured relational data such as graphs and knowledge bases. Recent work has increasingly focused on integrating graph-based representations and retrieval mechanisms with LLMs, giving rise to hybrid approaches that combine Graph Neural Networks (GNNs), Retrieval-Augmented Generation (RAG), and prompting strategies. This report surveys and synthesizes three representative approaches in this emerging area: *GNN-RAG*, *G-Retriever*, and *Large Language Models are Good Relational Learners*. Rather than reviewing these works independently, the report compares their methodological choices, retrieval strategies, prompting mechanisms, datasets, and evaluation protocols. The survey also highlights common strengths, limitations, and open challenges and motivates the broader design space of retrieval-augmented language reasoning over graphs.

1 Introduction and Motivation

Large Language Models (LLMs) have become the dominant paradigm in modern Natural Language Processing (NLP), achieving impressive results in tasks such as question answering, summarization, and reasoning. Despite their success, LLMs operate primarily over unstructured or weakly structured text and often struggle with tasks that require precise reasoning over explicit relational structure. In particular, reasoning over graphs - such as knowledge graphs, citation networks, or relational databases - poses challenges related to multi-hop dependencies and factual grounding.

From the perspective of graph-based learning, these challenges are well understood: relational information is inherently structured, sparse, and non-Euclidean, and effective reasoning often depends on identifying relevant local neighborhoods and multi-hop paths within large graphs. Graph Neural Networks (GNNs) have emerged as a principled approach for encoding such structure by propagating information along edges and learning representations that capture both local and higher-order connectivity patterns. However, while GNNs excel at structural representation learning, they lack the expressive language understanding and generative capabilities required for natural language reasoning and explanation.

To address these complementary limitations, recent research has explored hybrid systems that combine LLMs with external retrieval mechanisms and structured graph representations (GNN). Retrieval-Augmented Generation (RAG) has emerged as a general framework in which an LLM is conditioned on retrieved contextual evidence, improving factual accuracy and reducing hallucination. When the underlying data is graph-structured, however, the question of *what* to retrieve and *how* to represent it becomes non-trivial. Naively linearizing large graphs into text is inefficient and often ineffective, while purely neural graph embeddings may obscure explicit relational structure and reduce interpretability.

This report surveys three recent works that address these challenges by integrating graph-based retrieval and representation learning with LLM-based reasoning. The selected papers

- *GNN-RAG*, *G-Retriever*, and *Large Language Models are Good Relational Learners* - are conceptually aligned yet adopt distinct methodological perspectives, ranging from graph-driven retrieval and subgraph selection to soft graph prompting and relational conditioning of language models. Together, they illustrate the evolving landscape of hybrid NLP and graph-based systems that leverage both structural inductive biases and language modeling capabilities to support robust, multi-hop reasoning over complex relational data.

2 Background and Technical Foundations

2.1 Graph-Based Representation Learning

Graphs provide a natural abstraction for representing relational data, where nodes correspond to entities and edges encode relationships between them. In NLP-relevant settings, graphs often arise as knowledge graphs, citation networks, or textual graphs constructed from documents and entities. A key challenge in graph-based reasoning is identifying which parts of a large graph are relevant to a given query, particularly when reasoning requires multiple hops across the graph structure.

2.2 Graph Neural Networks

Graph Neural Networks (GNNs) are a class of models designed to learn representations over graph-structured data by iteratively aggregating information from neighboring nodes. Through message passing, GNNs can encode local and multi-hop structural information into continuous embeddings. In the context of language reasoning systems, GNNs are often used not as end-to-end predictors, but as intermediate components that encode relational structure or guide retrieval.

2.3 Large Language Models and Retrieval-Augmented Generation

LLMs model the probability of text sequences and can perform complex reasoning when provided with sufficient context. Retrieval-Augmented Generation (RAG) enhances LLMs by conditioning them on external evidence retrieved at inference time. In graph-centric settings, RAG raises specific questions regarding retrieval granularity, representation format, and prompt construction, all of which directly influence downstream reasoning performance.

3 Problem Setting: Reasoning Over Graph-Structured Data

Graphs provide a natural and expressive representation for structured data in which entities are connected through explicit relationships. As a result, reasoning over graphs has long been a central problem in machine learning and therefore crucial in domains such as knowledge and text graphs, social networks, and relational databases, where meaningful information is inherently distributed across interconnected components.

Within this broader context, the surveyed literature addresses two closely related but distinct problem formulations. The first concerns question answering and reasoning over graph-structured data using language models. Given a natural language query, the system must identify relevant portions of a graph and generate an accurate and coherent textual response. This problem is inherently multi-hop, as correct answers may depend on indirect relationships between entities, requiring the information from across multiple nodes and edges. Approaches such as [Mavromatis and Karypis, 2024] and [He et al., 2024] focus on this setting, where graph retrieval and subgraph selection play a central role in supporting language-based reasoning.

A second, complementary problem formulation focuses on learning and prediction over relational data, as exemplified by relational databases. Here, the goal is not to answer a single

query, but to perform predictive tasks such as classification or regression over entities whose attributes and behaviors are shaped by complex relational and temporal dependencies. In this setting, graph structure encodes inductive biases that are critical for generalization, and effective learning depends on capturing local neighborhoods, typed relations, and evolving interaction patterns. The Rel-LLM framework from [Wu et al., 2025] addresses this problem by integrating graph-based representation learning with large language models, enabling structured relational information to inform downstream predictions.

Across both problem settings, common challenges arise related to the selection of relevant portions of a graph and the preservation of structural dependencies. Graphs in real life scenarios are often large, sparse, heterogeneous, and dynamic, making it impractical to process them in their entirety. Instead, effective reasoning and learning depend on extracting compact, task-relevant subgraphs that retain essential relational information while excluding irrelevant or noisy connections. Graph Neural Networks provide a principled mechanism for encoding such subgraphs, as they are designed to model local neighborhoods and multi-hop connectivity through message passing and aggregation.

From an NLP perspective, an additional challenge lies in overcoming hallucinations and bridging the representational gap between structured graphs and sequential language models. Key design questions include how retrieved subgraphs should be represented for LLM consumption - whether through textual linearization, learned embeddings, or hybrid prompting strategies - and how to evaluate reasoning or prediction quality independently of retrieval performance. Together, these considerations define a unified problem setting at the intersection of language understanding and graph-based learning, which forms the foundation for the methodologies surveyed in this report.

4 Core Literature Review: Methodologies, Datasets and Benchmark evaluations

4.1 GNN-RAG: Graph Neural Retrieval for LLM Reasoning

GNN-RAG introduces a retrieval-augmented reasoning framework in which Graph Neural Networks are used explicitly to perform structured retrieval over a knowledge graph, while a large language model is responsible for natural language reasoning and answer generation. The key design principle of the approach is a strict separation between graph-based retrieval and language-based inference. Given a natural language question, the system first maps the query to entities in the knowledge graph and then applies a GNN-based retriever to score candidate answer nodes. For each candidate, the retriever identifies the shortest, high-confidence reasoning paths that connect the query entities to potential answers, thereby explicitly modeling multi-hop relational dependencies.

These retrieved reasoning paths are treated as a subgraph and the primary evidence for downstream inference. Rather than requiring the LLM to reconstruct graph structure implicitly, GNN-RAG verbalizes the subgraph into a concise textual form and supplies them to the language model as contextual input. This design allows the LLM to focus on compositional reasoning over graph-grounded evidence, while avoiding the need to process large portions of the graph or infer relational structure from unstructured text alone. The GNN component is trained to optimize retrieval quality, ensuring that the selected paths are both relevant to the query and structurally coherent, which is critical for supporting accurate multi-hop reasoning.

The effectiveness of GNN-RAG is evaluated on established knowledge graph question answering benchmarks, most notably WebQSP and Complex WebQuestions (CWQ), which are derived from Freebase and designed to test multi-hop and compositional reasoning. These datasets require systems to traverse multiple relations and combine evidence from several entities in order to produce correct answers. Experimental results demonstrate that GNN-RAG

consistently outperforms language-only baselines and text-based retrieval approaches, with particularly strong gains on CWQ, where questions frequently involve complex logical constraints. These findings highlight the importance of explicit graph-based retrieval for scalable and reliable reasoning in knowledge-intensive NLP tasks [Mavromatis and Karypis, 2024].

4.2 G-Retriever: Retrieval-Augmented Generation for Textual Graph Understanding

G-Retriever extends the RAG paradigm to general textual graphs, where nodes and edges are associated with rich natural language descriptions. Unlike the previously mentioned approach [Mavromatis and Karypis, 2024], that assumes canonical knowledge graph schemas, G-Retriever operates over heterogeneous graphs derived from text. The system retrieves a compact, connected subgraph relevant to the query using graph-aware retrieval and optimization techniques - specifically, a Prize-Collecting Steiner Tree (PCST) formulation - and then conditions an LLM on both a textualized version of the subgraph and learned graph embeddings.

From a graph learning perspective, G-Retriever explicitly addresses the challenges posed by large, sparse, and heterogeneous graphs, where relevant information is often distributed across loosely connected neighborhoods. By framing subgraph selection as an optimization problem, the method balances relevance and connectivity, ensuring that the retrieved evidence forms a coherent relational structure rather than a set of isolated nodes or edges.

From an NLP perspective, G-Retriever emphasizes prompt construction - combining *soft prompts* derived from graph embeddings with textual prompts obtained via graph linearization - and careful context management. By selectively retrieving and linearizing only the most relevant subgraph, the approach mitigates context length limitations, improves scalability to large graphs, and significantly reduces hallucination compared to text-only or embedding-only baselines.

The evaluation of G-Retriever is carried out on a diverse set of benchmarks designed to assess question answering over large textual graphs, most notably the GraphQA benchmark introduced in the paper. GraphQA spans multiple domains, including commonsense reasoning (ExplaGraphs), scene graph understanding (SceneGraphs), and knowledge graph question answering (WebQSP), thereby testing the model’s ability to operate across heterogeneous graph types with rich textual annotations. Despite substantial differences in graph size and structure, these datasets share the challenge that relevant evidence must be selectively retrieved to support multi-hop reasoning. Experimental results show that G-Retriever consistently outperforms text-only RAG baselines and naive graph linearization approaches, highlighting the importance of explicit subgraph retrieval and structure-aware context selection for scalable and faithful language-based reasoning over textual graphs [He et al., 2024].

4.3 Large Language Models as Relational Learners

Large Language Models are Good Relational Learners introduces *Rel-LLM*, a framework designed to enable large language models to perform learning and prediction over relational databases by explicitly incorporating graph-structured relational information. The central challenge addressed by the paper is that relational data - such as tables connected through foreign keys and temporal interactions - cannot be effectively represented through naive text linearization without losing structure or exceeding context limits. Rel-LLM addresses this by modeling relational databases as heterogeneous graphs and using a Graph Neural Network to extract compact, task-relevant subgraphs centered around target entities.

In Rel-LLM, the GNN encoder operates over these relational graphs to capture multi-hop dependencies, typed relations, and temporal structure, producing dense representations of local subgraphs that summarize the relational context relevant to a given prediction task. Rather than converting these subgraphs into text, the resulting representations are projected directly

into the latent space of the language model as *soft graph prompts*. This mechanism allows the LLM to condition its predictions on structured relational information without modifying the model architecture or relying on long textual prompts. A self-supervised masked attribute prediction objective is used to align graph representations with the language model’s embedding space, enabling effective integration of graph and language components.

The proposed approach is evaluated on RELBENCH, a benchmark suite constructed from real-world relational databases that include heterogeneous entity types, multi-relational edges, and temporal dynamics. RELBENCH comprises both classification and regression tasks that require models to reason over relational and temporal dependencies rather than isolated attributes. Experimental results show that Rel-LLM consistently outperforms traditional relational learning methods as well as LLM-based in-context learning baselines, particularly on tasks requiring multi-hop and temporal reasoning. These findings demonstrate that explicit graph-based conditioning, when integrated through soft prompting, provides a scalable and effective solution for learning and inference over complex relational data [Wu et al., 2025].

5 Discussion: Limitations, Open Challenges and Future Improvements

Across the surveyed works, the main failure mode is not that graph-aware RAG is inherently inefficient, but that retrieval quality and representation choices remain the dominant bottlenecks. In GNN-RAG, the authors explicitly analyze when GNN-based retrieval helps and when it fails: deep GNN retrieval is effective for multi-hop questions, but it can be weaker on simple (single-hop) queries where correct question–relation matching is more important than deep graph exploration. In these cases, an LLM-based retriever can select the right relation paths more reliably due to stronger natural-language understanding, motivating retrieval augmentation that unions GNN- and LLM-retrieved paths [Mavromatis and Karypis, 2024].

For G-Retriever, a key open issue is that even with subgraph retrieval and graph prompt tuning, answer faithfulness depends on whether the retrieved subgraph actually contains the necessary evidence. The authors show that retrieval substantially reduces hallucination compared to a baseline graph-prompt-tuned LLM, but hallucinations are not eliminated. A remaining challenge is guaranteeing that the produced answer (and any cited supporting nodes/edges) is fully grounded in the retrieved graph context. Also, the performance is sensitive to removing either the graph encoder or the textualized subgraph, suggesting that robust systems must jointly optimize *what* is retrieved and *how* it is presented to the LLM rather than relying on a single modality of graph-to-text conversion [He et al., 2024].

Finally, Rel-LLM highlights limitations that emerge when using LLMs for *relational prediction* rather than only question answering. In particular, the paper shows that model choice plays a critical role: LLMs with stronger pretrained relational knowledge and reasoning capabilities can substantially reduce prompt sensitivity and improve prediction reliability, indicating that architectural differences directly affect how well relational structure can be exploited [Wu et al., 2025].

6 Implications for Project Direction

The surveyed literature collectively motivates a broad and flexible design space for retrieval-augmented language reasoning over graph-structured data. Despite differences in architectural choices and application domains, the reviewed approaches share a common underlying paradigm: relevant subgraphs are retrieved from a larger graph and provided as contextual evidence to a large language model, which performs the final reasoning and answer generation. This paradigm highlights the importance of structured retrieval as a prerequisite for reliable language-based

inference in relational settings.

These works suggest multiple axes along which such systems can be analyzed and compared, including the choice of graph encoders for representing relational structure, retrieval and building strategies for selecting task-relevant subgraphs, the selection of language models for reasoning and generation, and the design of prompting mechanisms - ranging from explicit textual prompts to learned or soft graph prompts. Furthermore, evaluation practices vary across datasets and benchmarks, underscoring the need for systematic analysis of how retrieval quality, representation format, and prompting strategies jointly affect reasoning performance.

Taken together, the literature points toward the potential value of a more general framework for graph-aware retrieval-augmented generation, one that abstracts over specific datasets and task formulations. Such a framework would aim to support reasoning over diverse graph types by modularizing graph encoding, subgraph retrieval, and language-based inference, thereby facilitating generalization across domains while preserving the benefits of structured grounding. Rather than prescribing a single solution, the reviewed works collectively justify continued exploration of this design space and motivate further investigation into principled, dataset-agnostic approaches for integrating graph structure with language models.

7 Conclusion

This report surveyed three representative approaches at the intersection of NLP, graph learning, and retrieval-augmented generation. By comparing their methodologies, datasets, and evaluation strategies, the report highlighted both the promise and the challenges of integrating graph structure into language reasoning systems. As LLMs continue to evolve, hybrid approaches that combine retrieval, structure, and language are likely to play an increasingly important role in advancing robust and interpretable NLP systems.

References

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